

Supplementary Material

Table S1. Primary comparison

These three rows are consolidated into main-text Table 2.

Comparison	GSE70362 ECM up	GSE23130 ECM up	ECM gene concordance	Two-sided sign p
All available AF/LCM samples	46.7%	87.1%	40.0%	0.3616
Strict grade I-IV AF/LCM, adjusted	46.7%	77.4%	23.3%	0.0052
Strict grade I-IV AF/LCM, grade only	50.0%	74.2%	23.3%	0.0052

Table S2. Cohort effect diagnostics

Analysis set	Samples	Donors	Genes	Up	Median effect	Global shift
GSE70362 AF, all grades	24	18	18479	47.6%	-0.0027	False
GSE70362 AF, grade I-IV	20	16	18479	43.6%	-0.0089	False
GSE23130, all processing	23	NA	22107	49.7%	-0.0002	False
GSE23130 LCM, grade I-IV	15	NA	22107	55.0%	+0.0075	False
GSE70362 AF, grade I-IV, grade only	20	16	18479	43.4%	-0.0084	False
GSE23130 LCM, grade I-IV, grade only	15	NA	22107	51.6%	+0.0021	False

Table S3. Internal split-half reliability

cohort	gene_space	median_spearman	n_splits
GSE70362_AF_all	all	0.1983099325567557	100

cohort	gene_space	median_spearman	n_splits
GSE70362_AF_all	ECM	0.4075639599555061	100
GSE70362_AF_I_IV	all	0.1383736790830905	100
GSE70362_AF_I_IV	ECM	0.3690767519466074	100
GSE23130_all	all	-0.0654864000493762	100
GSE23130_all	ECM	0.044758064516129	100
GSE23130_LCM_all	all	0.1337130096184392	100
GSE23130_LCM_all	ECM	0.4084677419354838	100
GSE70362_AF_I_IV_grade_only	all	0.2353076937601182	100
GSE70362_AF_I_IV_grade_only	ECM	0.3388209121245829	100
GSE23130_LCM_grade_only	all	0.2382855306866113	100
GSE23130_LCM_grade_only	ECM	0.7268145161290323	100

Table S4. Full MSigDB audit

The complete audit of 1554 modules is provided as `results/msigdb\_module\_audit.csv`.

Table S5. Pairwise tissue cohort concordance

Pair	Genes	Concordance	Exact p
GSE186542 vs GSE167199	13441	58.6%	5.673517 x 10 ⁻⁹⁰
GSE167199 vs GSE146904	13788	57.4%	1.948720 x 10 ⁻⁶⁷
GSE23130 vs GSE146904	13960	53.6%	4.187344 x 10 ⁻¹⁸
GSE70362 vs GSE23130	14004	51.4%	3.510011 x 10 ⁻⁴
GSE23130 vs GSE207176	14004	51.3%	7.647001 x 10 ⁻⁴
GSE23130 vs GSE167199	13827	50.9%	0.0135

Pair	Genes	Concordance	Exact p
GSE70362 vs GSE186542	13563	49.7%	0.7593
GSE70362 vs GSE146904	13960	49.4%	0.9212
GSE70362 vs GSE207176	14004	49.1%	0.9880
GSE23130 vs GSE186542	13563	48.7%	0.9988
GSE70362 vs GSE167199	13827	47.6%	1.0000
GSE186542 vs GSE146904	13554	44.2%	1.0000
GSE186542 vs GSE207176	13563	42.5%	1.0000
GSE146904 vs GSE207176	13960	41.8%	1.0000
GSE167199 vs GSE207176	13827	38.7%	1.0000

Table S6. Prespecified mechanism modules

The complete comparison is provided as `results/curated\_module\_audit.csv`.

Table S7. Donor-level sensitivity

Sensitivity model	Replicates	Fraction up	Up 2.5%	Up 97.5%	Concordance	Conc 2.5%	Conc 97.5%	Effect rho
Donor cluster bootstrap	300	50.0%	36.7%	60.0%	33.3%	20.0%	53.3%	-0.450
One sample per donor	300	50.0%	46.7%	53.3%	30.0%	23.3%	33.3%	-0.498

Table S8. Reliability attenuation

Latent correlation	Expected concordance	2.5%	97.5%
0.0	50.0%	33.3%	66.7%
0.2	53.5%	36.7%	70.0%
0.4	57.1%	40.0%	73.3%

Latent correlation	Expected concordance	2.5%	97.5%
0.6	60.8%	43.3%	76.7%
0.8	64.6%	46.7%	80.0%

Table S9. Effective independent genes

cohort	gene_space	effective_genes
GSE70362_AF_I_IV	ECM	6.842470672703271
GSE23130_LCM_all	ECM	5.9055599648777015

Table S10. Expression-matched ECM null

module	n_genes	empirical_p_greater	empirical_p_less	empirical_p_two_sided	null_median	null_upper_95
ECM remodelling	30	0.9988002399520096	0.0045990801839632	0.0091981603679264	0.5	0.6333333333333333

Table S11. Cross-half positive controls

Cohort	Replicates	Concordance	2.5%	97.5%	Effect rho
GSE70362_AF_I_IV	300	63.3%	46.7%	80.0%	0.365
GSE23130_LCM_all	300	64.5%	38.7%	77.4%	0.380

Table S12. Weighted concordance

Metric	Value
Unweighted sign concordance	23.3%
Effect-size weighted concordance	24.2%
Top-half effect concordance	26.7%

Metric	Value
Effect-size Spearman rho	-0.495

Table S13. Power calibration

Latent correlation	Mean concordance	Power, nominal n	Power, effective n	Effective genes
0.0	50.0%	4.4%	0.0%	5
0.2	53.7%	6.2%	0.0%	5
0.4	57.4%	11.1%	0.0%	5
0.6	60.9%	20.4%	0.0%	5

Table S14. Module score slopes

Cohort	Genes	Score slope
GSE70362_AF_I_IV	30	-0.0185
GSE23130_LCM_all	31	+0.1729

Table S15. Permutation-method sensitivity

Method	GSE70362 p	GSE23130 p
stratified	0.7500	0.1000
freedman_lane	0.3700	0.0700
rotation	0.6600	0.0100

Table S16. Exact-contrast cohort pairs

Pair	Genes	Concordance
GSE186542 vs GSE167199	13441	58.6%

Table S17. Compartment-matched cohort pairs

Pair	Genes	Concordance
GSE70362 vs GSE23130	14004	51.4%
GSE186542 vs GSE207176	13563	42.5%
GSE167199 vs GSE207176	13827	38.7%

Table S18. GSE176205 NP directional validation

Module	Genes	GSE176205 up	GSE176205 median	GSE70362 NP up	GSE70362 median	Concordance	Effect rho
Inflammation/ cytokine	27	44.4%	-0.222	60.0%	+0.010	59.3%	+0.031
Angiogenesis/ vascular	17	41.2%	-0.424	72.2%	+0.036	58.8%	-0.029
Macrophage/ myeloid	18	50.0%	+0.066	68.4%	+0.016	55.6%	+0.480
Immune activation	30	43.3%	-0.058	51.6%	+0.004	53.3%	+0.058
ECM remodelling	29	16.7%	-0.557	56.7%	+0.007	44.8%	+0.231
Apoptosis/ autophagy	26	34.6%	-0.392	50.0%	+0.005	38.5%	-0.117
Adhesion/ cytoskeleton	23	58.3%	+0.265	39.1%	-0.016	34.8%	-0.139

Table S19. GSE17077 paired senescence analysis

Module	Genes	Senescent up	Median effect	One-sided p
Angiogenesis/vascular	18	44.4%	-0.022	0.7596588134765634
Adhesion/cytoskeleton	24	41.7%	-0.008	0.8462718725204479
Immune activation	31	41.9%	-0.006	0.8594792424701133

Module	Genes	Senescent up	Median effect	One-sided p
ECM remodelling	31	41.9%	-0.017	0.8594792424701133
Apoptosis/autophagy	25	40.0%	-0.023	0.8852385282516482
Inflammation/cytokine	30	40.0%	-0.010	0.8997557889670215
Macrophage/myeloid	20	35.0%	-0.008	0.9423408508300802

Table S20. Third-cohort registry summary

Tier	Datasets
B	2
C	14
core	2
exclude	16
unverified	5

Table S21. Literature scoping audit

Scoping metric	Articles
Open-access articles in scoping corpus	95
Articles using more than one GSE	79
Articles with consistency language	71
Multi-cohort articles with consistency language	62
Articles describing gene-level concordance	0
Articles mentioning permutation	9
Articles mentioning FDR	35
Articles mentioning power	35

Table S22. Reproduced blood cohorts

Cohort	Design	Samples
GSE124272	8 LDH vs 8 healthy	16
GSE150408	42 IDD vs 17 healthy; 25 treatment	59
GSE150408 untreated	17 untreated IDD vs 17 healthy	34

Table S23. Probe-to-symbol collapsing sensitivity

These rows are consolidated into main-text Table 2.

Collapse rule	GSE70362 ECM up	GSE23130 ECM up	ECM concordance	Two-sided sign p
Highest-mean probe (primary)	46.7%	77.4%	23.3%	0.0052
First probe	46.7%	83.9%	30.0%	0.0428
Median probe	46.7%	74.2%	33.3%	0.0987
Random probe, median (2.5-97.5%)	46.7%	71.0% (61.3%-80.6%)	36.7% (23.3%-48.4%)	0.2005 (median)

The primary analysis used the highest-mean probe per gene. Alternative collapsing rules and 100 random-probe selections did not exceed 50% ECM concordance, but they showed that the exact below-chance level in the primary analysis was sensitive to probe selection.

Table S24. Gene-identity sensitivity

Metric	Primary mapping	Alias-aware mapping
Platform gene symbols	22107	22107
Current canonical symbols	17704	17704
Alias-only symbols recovered	0	1098
Unresolved symbols	3305	3305
Ambiguous alias keys removed	4434	4434
Modules tested	1554	1572

Metric	Primary mapping	Alias-aware mapping
Significant in one cohort	673	692
Significant in both cohorts	31	31
Nominal gene-level results	14	15
Strict FDR results	0	0

The primary mapping retained supplied platform symbols. The alias-aware mapping additionally used current synonyms and nomenclature-authority symbols from human gene_info. It recovered 1,098 alias-only symbols but left 3,305 unresolved and removed 4,434 ambiguous alias keys. The strict FDR conclusion remained zero modules, while broader nominal counts changed modestly.

Table S25. Scale-invariant trend sensitivity

Analysis	GSE70362 ECM up	GSE23130 ECM up	ECM concordance	Two-sided sign p
Adjusted OLS (primary)	46.7%	77.4%	23.3%	0.0052
Unadjusted Spearman trend	46.7%	64.5%	33.3%	0.0987
Batch/source-stratified Spearman trend	53.3%	74.2%	36.7%	0.2005

Spearman trends are invariant to any strictly increasing transformation of each gene across samples and therefore test whether the direction call is driven by platform-specific scale assumptions. The stratified analysis averages gene-wise Spearman trends within batch or source strata with at least four samples and three observed grades.

Table S26. Module-family FDR and overlap sensitivity

Mapping	Collection	Modules	Nominal gene-level	Collection-local FDR
primary	Hallmark	50	0	0
primary	KEGG	182	3	0
primary	Reactome	1322	11	0
alias-aware	Hallmark	50	0	0
alias-aware	KEGG	184	3	0

Mapping	Collection	Modules	Nominal gene-level	Collection-local FDR
alias-aware	Reactome	1338	12	0

The module sets were not independent. Among 1,554 tested modules, 243,830 pairs had nonzero Jaccard overlap, 1,636 exceeded Jaccard 0.5 and 94 exceeded 0.8. Collection-local FDR was zero for Hallmark, KEGG and Reactome under both primary and alias-aware mappings.

Table S27. Nominally replicated modules

The 14 modules that reached Level 3 (candidate gene-level replication) at nominal p-values are a hypothesis-generating shortlist, not a biological conclusion; 0 survived the strict FDR gate. The top 10 by gene-level concordance are shown here; the full list is in `results/msigdb_module_audit.csv``.

Module	Genes	GSE70362 up	GSE23130 up	Concordance
KEGG_PROTEIN_EXPORT	22	95.5%	77.3%	81.8%
REACTOME_HSP90_CHAPERONE_CYCLE_FOR_STEROID_HORMONE_RECEPTORS_SHR_IN_THE_PRESENCE_OF_LIGAND	49	65.3%	77.6%	75.5%
REACTOME_ANTIGEN_PRESENTATION_FOLDING_ASSEMBLY_AND_PEPTIDE_LOADING_OF_CLASS_II_MHC	29	69.0%	75.9%	72.4%
REACTOME_CHAPERONE_MEDIATED_AUTOPHAGY	21	71.4%	71.4%	71.4%
REACTOME_RHO_GTPASES_ACTIVATE_IQGAPS	27	70.4%	85.2%	70.4%
REACTOME_SRP_DEPENDENT_COTRANSLATIONAL_PROTEIN_TARGETING_TO_MEMBRANE	63	77.8%	87.3%	68.3%
KEGG_PATHOGENIC_ESCHERICHIA_COLI_INFECTION	44	72.7%	77.3%	68.2%
REACTOME_COOPERATION_OF_PREFOLDIN_AND_TRIC_CCT_IN_ACTIN_AND_TUBULIN_FOLDING	31	67.7%	80.6%	67.7%

Module	Genes	GSE70362 up	GSE23130 up	Concordance
REACTOME_CYCLIN_D_ASSOCIATED_EVENTS_IN_G1	44	65.9%	63.6%	65.9%
KEGG_VIBRIO_CHOLERA_E_INFECTION	54	63.0%	74.1%	63.0%

Figure S1. Positive controls and power calibration

(A) Cross-cohort ECM concordance compared with within-cohort cross-half positive controls. (B) Power of the nominal 30-gene sign test across simulated latent correlations. (C) Unweighted, effect-size-weighted and top-half effect concordance. This figure was moved from the main text (Figure 6 in earlier drafts) to keep the main-text display-item count within the journal limit.

Technical diagnostic tables

The grade, batch, processing-method and source crosstabs are provided in `results/confound\_\*.csv`. These tables are part of the stopping rules: the unrestricted GSE23130 comparison is not used as the primary replication test because grade V occurs only in homogenized samples.